

Muntakimur Rahaman

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Research Interests: Post-Quantum Cryptography & Quantum-Safe Networking (PQC, QKD) · Security of Communication Protocols & Architectures · Encrypted-Traffic & Network-Security Analysis · AI for Cybersecurity · Resource-Constrained / Edge Security

EDUCATION

M.Eng.Sc. (by Research) in Cyber-Security & Networking

2025 – Present

Multimedia University (MMU), Faculty of Artificial Intelligence and Engineering

Cyberjaya, Malaysia

- Graduate Research Assistant. Research on AI-driven, quantum-safe security: data-efficient learning for encrypted-traffic classification, PQC/QKD orchestration, and adaptive intrusion detection.
- Expected completion: December 2026 (thesis submitted; available to start thereafter). One peer-reviewed journal paper published (see Publications).

B.Sc. (Hons.) in Computer Science and Engineering

2016 – 2020

Bangladesh Army International University of Science and Technology (BAIUST)

Comilla, Bangladesh

- CGPA: 3.74 / 4.00.
- Undergraduate thesis: “Bangla Song Genre Classification using Bi-LSTM” — deep sequence modelling for audio genre recognition.

RESEARCH EXPERIENCE

Graduate Research Assistant

2025 – Present

Faculty of Artificial Intelligence and Engineering, Multimedia University

Cyberjaya, Malaysia

- Designed a hybrid reinforcement-learning controller for the joint orchestration of Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD) in quantum-safe networks — directly relevant to embedding post-quantum security into communication infrastructure (Gold Medal, ITEX 2026).
- Led, under supervision, a PRISMA 2020 systematic review of data-efficient and adaptive learning for encrypted-traffic classification under modern protocols (TLS 1.3, QUIC, Tor, VPN), screening 500 studies down to 22; published as first author in *Computers* (MDPI).
- Designed *Cognitive Traffic Adaptation*, an episodic Transformer framework for cross-domain encrypted-traffic analysis targeting concept drift and zero-day attacks; filed as a patent application.
- Benchmarked deep and ensemble models for HTTPS traffic classification from packet-burst statistics, with emphasis on interpretability (accepted at an IEEE symposium; see Publications).

PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Journal Articles

- Rahaman, M.**, Mahmud, A., Abd Aziz, A., Abujawa, O. M. S., & Chin, J.-J. (2026). A Systematic Literature Review on Data-Efficient and Adaptive Learning Techniques for Encrypted Traffic Classification Under Modern Protocols. *Computers*, 15(5), 319. <https://doi.org/10.3390/computers15050319>

Conference Papers

- Rahaman, M.** et al. Benchmarking Deep and Ensemble Learning for HTTPS Traffic Classification: The Case for Packet-Burst Statistics and Interpretability. *IEEE 10th International Symposium, Penang, Malaysia 2026* (accepted).

PATENTS & INTELLECTUAL PROPERTY

- Rahaman, M.** (filed). “*Cognitive Traffic Adaptation: An Episodic Transformer Framework for Cross-Domain Encrypted Traffic Analysis*.” Patent application filed, 2026.

AWARDS & HONORS

Gold Medal — ITEX 2026 (International Invention, Innovation & Technology Exhibition)

2026

For “Hybrid RL-based PQC and QKD Orchestration for Quantum-Safe Networking” Kuala Lumpur, Malaysia

Cisco AI Innovation Award

2025

For an AI-powered railway safety system achieving >90% incident-detection accuracy

TECHNICAL SKILLS

Languages: Python, Rust, SQL/PostgreSQL, Java, ReactJS, Node.js

Security & Networking: Post-quantum cryptography (PQC) & QKD; encrypted-traffic analysis (TLS 1.3, QUIC, VPN, Tor); intrusion detection; communication-protocol & security-architecture analysis

Embedded & Edge: NVIDIA Jetson edge deployment, real-time on-device inference (resource-constrained computing)

Machine Learning & Deep Learning: TensorFlow, PyTorch, Scikit-learn; Transformers, CNNs, LSTMs; reinforcement learning; few-shot, zero-shot & meta-learning; ensemble methods (XGBoost, CatBoost, Random Forest); computer vision (OpenCV, YOLOv8); time-series (Prophet, SAX)

MLOps & Tools: Docker, Git, Airflow, MLflow, DVC, FastAPI, SageMaker

Cloud & Data: AWS, GCP, BigQuery, Spark/PySpark; Plotly/Dash, Mapbox, GeoPandas

GenAI & LLM: LangChain, LangGraph, RAG, OpenAI / GPT-4o, Gemini, Faiss, ChromaDB

RELEVANT PROFESSIONAL EXPERIENCE

GenAI & LLM Data Scientist (Part-time)

Jan 2025 – Mar 2025

VedTeq (Remote)

India

- Implemented three production-grade RAG pipelines (LangChain + Faiss), reducing average latency by 45 ms.
- Developed GPT-4o-mini-powered agents that cut human ticket load by 25%, and a Gemini-powered personalised chatbot for improved suggestions.
- Authored a modular RAG framework that reduced new-client onboarding from three weeks to five days, and built a profile-based university recommendation engine (98% accuracy).

Senior Data Scientist

Jun 2022 – Jan 2025

Innova Analytics (Remote)

Australia

- Re-architected the streaming stack (GCP Pub/Sub + PySpark) for a 30% throughput increase, and deployed an LLM-powered well-data explorer that shortened insight lead-time from hours to minutes.
- Extended pump run-life by 12% across 400 assets via predictive-maintenance models, and deployed a data-driven application giving real-time visibility into pressure, flow, and downtime under well shutdowns.
- Implemented YOLOv8 PPE detection (98% accuracy) on Jetson edge devices and an AI-powered railway safety system (>90% incident detection) — recognised with the Cisco AI Innovation Award (2025).

Data Scientist

Apr 2021 – May 2022

Leading Engineers Solutions (Remote)

Australia

- Developed a housing-price prediction API ($R^2 = 0.95$) serving over 50,000 monthly users.
- Automated council-document classification using BERT, achieving a 31-percentage-point accuracy uplift, and built Dash/Plotly geospatial dashboards that halved disaster-response map-preparation time.

CERTIFICATIONS

DeepLearning.AI TensorFlow Developer Professional Certificate · IBM Data Science Professional Certificate · IBM Spark — Level 1 · Generative AI Essentials on AWS

LEADERSHIP & PROFESSIONAL SERVICE

Volunteer & Event Management

2026

IEEE WCNC 2026, Kuala Lumpur Convention Centre (KLCC)

Kuala Lumpur, Malaysia